The `drell_yan_lo` Numerical Experiment

Computational HEP

1 Purpose of the experiment

This numerical experiment computes a deliberately simple version of Drell–Yan production at a proton–proton collider:

$$\mathrm{pp} \to Z \to \ell^+\ell^-.$$

The implementation is intentionally pedagogical. It does not try to be a precision Drell–Yan prediction. Instead, it is designed to expose the three main ingredients that appear in almost every hadron-collider calculation:

1. the proton is not an elementary particle, so the incoming hard partons carry only fractions x_1 and x_2 of the proton momenta;
2. the probability densities for finding those partons are the PDFs;
3. the short-distance partonic reaction is simple and calculable.

In this experiment the hard reaction is

$$q\bar{q} \to Z,$$

and the decay $Z \to \ell^+\ell^-$ is included only through a branching-ratio factor. Therefore the final state is treated as an on-shell object with mass $m_{\ell\ell}$, not as two resolved leptons with an angular distribution. This is why the phase space is a $2 \to 1$ phase space.

2 Hadron-collider kinematics

Let the two proton beams have centre-of-mass energy E_{cm}. The proton momenta are taken to be massless:

$$P_1^\mu = \frac{E_{\mathrm{cm}}}{2}(1, 0, 0, +1), \qquad P_2^\mu = \frac{E_{\mathrm{cm}}}{2}(1, 0, 0, -1).$$

The hard collision is not between the two full protons. It is between two partons inside the protons. If the parton from proton 1 carries momentum fraction x_1, and the parton from proton 2 carries momentum fraction x_2, then

$$p_1^\mu = x_1 P_1^\mu, \qquad p_2^\mu = x_2 P_2^\mu.$$

The partonic centre-of-mass energy is

$$\hat{s} = (p_1 + p_2)^2 = x_1 x_2 E_{\mathrm{cm}}^2.$$

For a $2 \to 1$ reaction producing a final-state object of mass $m_{\ell\ell}$, the partonic energy is fixed:

$$\hat{s} = m_{\ell\ell}^2.$$

It is useful to define

$$\tau = x_1 x_2 = \frac{m_{\ell\ell}^2}{E_{\mathrm{cm}}^2}.$$

The longitudinal boost of the partonic collision is described by the rapidity of the produced object,

$$y = \frac{1}{2}\log\frac{x_1}{x_2}.$$

Solving these two equations gives

$$x_1 = \sqrt{\tau}\, e^{+y}, \qquad x_2 = \sqrt{\tau}\, e^{-y}.$$

Because $0 < x_1 < 1$ and $0 < x_2 < 1$, the rapidity is bounded:

$$-\log\frac{E_{\mathrm{cm}}}{m_{\ell\ell}} \le y \le +\log\frac{E_{\mathrm{cm}}}{m_{\ell\ell}}.$$

The code does not rederive this mapping itself. It uses the existing `FlatPhaseSpace` generator in $2 \to 1$ proton-proton mode:

```
FlatPhaseSpace([0,0], [m_ll], beam_types=(1,1)).
```

For a $2 \to 1$ final state this generator fixes $\tau = m_{\ell\ell}^2/E_{\mathrm{cm}}^2$, samples only the rapidity variable, and returns x_1, x_2, and the Jacobian. This is important: the Drell–Yan experiment is not manually inventing a Bjorken-x mapping. It is using the same phase-space machinery as the other hadron-collider experiments.

3 PDFs and partonic luminosities

A PDF $f_i(x, \mu_F)$ answers the following question:

> At factorisation scale μ_F, how likely is it to find a parton of type i carrying a fraction x of the proton momentum?

The index i can be a gluon, a quark, or an antiquark. In Drell–Yan at leading order, only quark–antiquark initial states contribute:

$$q\bar{q} \to Z.$$

For a proton-proton collider, either beam can provide the quark and the other beam can provide the antiquark. Therefore, for each quark flavour q, the partonic luminosity is the sum of two beam assignments:

$$\mathcal{L}_{q\bar{q}}(x_1, x_2, \mu_F) = f_q(x_1, \mu_F) f_{\bar{q}}(x_2, \mu_F) + f_{\bar{q}}(x_1, \mu_F) f_q(x_2, \mu_F).$$

The code keeps these two terms separately in the output table, because it is instructive to see that they are generally not equal away from $y = 0$. At positive rapidity, $x_1 > x_2$; at negative rapidity, $x_2 > x_1$. Since quark and antiquark PDFs are different functions, the two beam assignments probe different regions of the proton structure.

LHAPDF returns $x f_i(x, \mu_F)$, not $f_i(x, \mu_F)$. The experiment therefore converts it back by dividing by x:

$$f_i(x, \mu_F) = \frac{x f_i(x, \mu_F)}{x}.$$

This division is explicit in the code because it is one of the easiest places to make a factor-of-x mistake.

4 The Z-boson couplings

The neutral-current $Zf\bar{f}$ vertex is written in the convention

$$i\frac{e}{2s_wc_w}\gamma^\mu(v_f - a_f\gamma_5).$$

Here e is the electromagnetic coupling, and

$$s_w = \sin\theta_w, \qquad c_w = \cos\theta_w$$

are the sine and cosine of the weak mixing angle. The dimensionless vector and axial couplings are

$$v_f = T_f^3 - 2Q_f s_w^2, \qquad a_f = T_f^3.$$

Q_f is the electric charge in units of the proton charge. T_f^3 is the third component of weak isospin. For example,

$$(Q_u, T_u^3) = \left(+\frac{2}{3}, +\frac{1}{2}\right), \qquad (Q_d, T_d^3) = \left(-\frac{1}{3}, -\frac{1}{2}\right).$$

The code stores these charge and isospin assignments for d, u, s, c, b quarks. For a chosen flavour q, it computes v_q and a_q, and then weights the channel by $v_q^2 + a_q^2$.

5 The hard matrix element

The partonic process is

$$q(p_1) + \bar{q}(p_2) \to Z(k), \qquad k^2 = m_{\ell\ell}^2.$$

The amplitude is

$$\mathcal{M} = \bar{v}(p_2)\left[i\frac{e}{2s_wc_w}\gamma^\mu(v_q - a_q\gamma_5)\right]u(p_1)\,\epsilon_\mu^*(k).$$

The experiment uses the spin- and colour-averaged squared matrix element

$$\overline{|\mathcal{M}|^2} = \frac{g_Z^2\,\hat{s}}{12}\left(v_q^2 + a_q^2\right), \qquad g_Z = \frac{e}{s_wc_w}.$$

Let us unpack this formula.

- The bar over $|\mathcal{M}|^2$ means that we average over the unobserved spin and colour states of the incoming quark and antiquark.
- The factor $1/4$ from spin averaging is already included.
- The factor $1/3$ from colour averaging is already included. The final Z is colourless, so only a colour-singlet quark-antiquark combination contributes.
- The interference between vector and axial terms vanishes after the spin sum for this fully inclusive $2 \to 1$ production rate.

This is what is meant by "the LO matrix element is trivial" in this numerical experiment: the code does not call a generated matrix-element library. It uses this one analytic expression directly.

6 The $2 \to 1$ partonic cross section

For a $2 \to 1$ process, the partonic cross section contains a delta function that enforces the on-shell condition:

$$\mathrm{d}\hat{\sigma}_q = \frac{\pi}{\hat{s}}\overline{|\mathcal{M}_q|^2}\delta(\hat{s} - m_{\ell\ell}^2).$$

This expression is compact, so it is worth saying what it means. The incoming partons can have many possible x_1, x_2, but only combinations with

$$x_1 x_2 E_{\mathrm{cm}}^2 = m_{\ell\ell}^2$$

can produce an on-shell object of mass $m_{\ell\ell}$. The delta function removes one integration variable. In the code, the phase-space generator handles this for us. In its $2 \to 1$ mode it fixes $\tau = m_{\ell\ell}^2/E_{\mathrm{cm}}^2$ and leaves only the rapidity integral.

The hadronic cross section in this approximation is therefore

$$\sigma(\mathrm{pp} \to Z \to \ell^+\ell^-) = \sum_q \int \mathrm{d}x_1\,\mathrm{d}x_2\,\mathcal{L}_{q\bar{q}}(x_1, x_2, \mu_F)\,\frac{\pi}{\hat{s}}\overline{|\mathcal{M}_q|^2}\delta(\hat{s} - m_{\ell\ell}^2)\,\mathrm{BR}(Z \to \ell^+\ell^-).$$

Equivalently, after using τ and y,

$$\sigma = \sum_q \int_{-Y}^{+Y} \mathrm{d}y\,\frac{1}{E_{\mathrm{cm}}^2}\,\mathcal{L}_{q\bar{q}}(x_1(y), x_2(y), \mu_F)\,\pi\,\frac{\overline{|\mathcal{M}_q|^2}}{m_{\ell\ell}^2}\,\mathrm{BR}(Z \to \ell^+\ell^-),$$

where

$$Y = \log\frac{E_{\mathrm{cm}}}{m_{\ell\ell}}.$$

The factor $1/E_{\mathrm{cm}}^2$ is the Jacobian from the delta function $\delta(x_1 x_2 E_{\mathrm{cm}}^2 - m_{\ell\ell}^2)$. In the code this is part of the `FlatPhaseSpace` Jacobian.

7 The branching ratio

By default the experiment returns a cross section for

$$\mathrm{pp} \to Z \to \ell^+\ell^-.$$

The branching ratio is estimated at tree level. For one charged lepton flavour,

$$\Gamma(Z \to \ell^+\ell^-) = \frac{G_F m_Z^3}{6\sqrt{2}\pi}\left(v_\ell^2 + a_\ell^2\right),$$

and

$$\mathrm{BR}(Z \to \ell^+\ell^-) = \frac{\Gamma(Z \to \ell^+\ell^-)}{\Gamma_Z}.$$

Here Γ_Z is taken from the model parameters. If one wants inclusive on-shell Z production rather than a charged-lepton final state, one can run with

`--branching_ratio 1`.

8 Monte Carlo integration

The integrator samples a point $u \in [0, 1]$. The phase-space generator maps this to a rapidity

$$y(u) = -Y + 2Yu,$$

and then to

$$x_1(u) = \sqrt{\tau}\, e^{y(u)}, \qquad x_2(u) = \sqrt{\tau}\, e^{-y(u)}.$$

At that point the code computes

$$w(u) = J(u) \sum_q \left[f_q(x_1, \mu_F) f_{\bar{q}}(x_2, \mu_F) + f_{\bar{q}}(x_1, \mu_F) f_q(x_2, \mu_F)\right] \pi \frac{\overline{|\mathcal{M}_q|^2}}{m_{\ell\ell}^2} \mathrm{BR}(Z \to \ell^+\ell^-),$$

where $J(u)$ is the phase-space/Bjorken-x Jacobian returned by the generator. The unit conversion

$$1\ \mathrm{GeV}^{-2} = 0.389379338 \times 10^9\ \mathrm{pb}$$

is applied at the end.

The integral over u gives the total cross section:

$$\sigma = \int_0^1 \mathrm{d}u\, w(u).$$

The numerical integrator reports an estimate and an integration error after each iteration.

9 Rapidity distribution

In addition to the Monte Carlo integral, the experiment writes a simple rapidity table. This table is evaluated deterministically at bin centres. Since the generator weight is a weight per unit u, and

$$\mathrm{d}y = 2Y\, \mathrm{d}u,$$

the code divides by the rapidity range $2Y$ to obtain

$$\frac{\mathrm{d}\sigma}{\mathrm{d}y}.$$

The output files are

`drell_yan_lo_rapidity.csv`, `drell_yan_lo_rapidity.png`.

The CSV table also includes separate channel contributions such as `u qbar` and `ubar q`. These labels indicate which beam provided the quark and which beam provided the antiquark.

10 Example commands

A default run near the Z pole is

```
./run.py drell_yan_lo -ni 10 -npi 1000 \
  --e_cm 13000 \
  --quark_flavours d,u,s,c \
  --lepton mu \
  --output_dir drell_yan_lo_output
```

For inclusive on-shell Z production, omit the branching-ratio suppression:

```
./run.py drell_yan_lo -ni 10 -npi 1000 \
  --branching_ratio 1 \
  --output_dir drell_yan_lo_inclusive
```

11 What is deliberately not included

This experiment is intentionally minimal. The following physics effects are not included:

- photon exchange, $\gamma^* \to \ell^+\ell^-$;
- γ-Z interference;
- the finite-width Breit-Wigner line shape of the Z;
- the explicit two-lepton decay angles;
- QCD radiation such as $q\bar{q} \to Zg$ or $qg \to Zq$;
- virtual loop corrections;
- detector cuts on the charged leptons.

These omissions are not accidents. The purpose of the experiment is to show, as cleanly as possible, how a hadron-collider observable is built from partonic luminosities and a hard matrix element.